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Attenuation of Pressure Swings Along the Endotracheal Tube Is Indicative of Optimal Distending Pressure During High-Frequency Oscillatory Ventilation in a Model of Acute Lung Injury

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Summary. We tested the hypothesis that during high-frequency oscillatory ventilation, the oscillatory pressure ratio (OPR) is minimal at the optimal mean airway pressure (Paw). OPR is defined as the ratio of pressure swings at the distal end and the proximal opening of the endotracheal tube. Optimal Paw was assumed to be the lowest Paw at which the physiological shunt fraction was below 0.1. Acute lung injury was produced by saline lung lavage of pigs who were then subjected to a stepwise increase of Paw to impose underinflation, optimal inflation, and overdistention (inflation phase), followed by a stepwise decrease of Paw (deflation phase).

OPR reached a minimum of 0.10 ± 0.01 at Paw $= 31 \pm 4$ cm H_2O during the inflation phase and a minimum of 0.04 ± 0.01 at Paw $= 18 \pm 1$ cm H_2O during the deflation phase. Optimal Paw was 31 ± 4 cm H_2O on the inflation limb and 14 ± 2 cm H_2O on the deflation limb. Paw at the minimal OPR was not significantly different from the optimal Paw during the inflation phase, and slightly but significantly higher (4.1 ± 1.6 cm H_2O) during the deflation phase.

In conclusion, a consistent relationship was found between OPR and Paw, with a minimum in all animals. The minimal OPR coincides fairly well with the Paw where oxygenation is optimal. **Pediatr Pulmonol. 2002; 33:429–436.** © 2002 Wiley-Liss, Inc.

Key words: acute lung injury; high-frequency oscillatory ventilation; tracheal pressure; endotracheal tube; gas exchange.

INTRODUCTION

In newborn infants suffering from idiopathic respiratory distress syndrome (IRDS), high-frequency oscillatory ventilation (HFOV) is frequently used as a mode of respiratory support.[1] IRDS is caused by a lack of lung surfactant, resulting in alveolar collapse.[2] HFOV opens atelectatic lung regions by maintaining appropriate mean airway pressure (Paw), while the superimposed small oscillations of air flow provide gas exchange.[3] It has been suggested that low Paw levels can adversely affect patient outcome,[4] and excessive Paw levels can lead to alveolar overdistention and barotrauma.[5] Therefore, it is crucial to find the safe window of Paw in every patient.[5] According to the ''open lung concept,''[6,7] the optimal ventilatory condition during conventional mechanical ventilation (CMV) is the lowest positive end-expiratory pressure (PEEP) to achieve a fully recruited lung, i.e., at a physiological shunt fraction (Q'_s/Q'_T) below 0.1. Determination of Q'_s/Q'_T, however, requires invasive access for mixed venous sampling, not commonly used in infants. Measurement of respiratory mechanics has also been suggested as a way to find the optimal ventilatory condition.[8] However, it was shown that respiratory system

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compliance (C_{rs}), measured using the single-occlusion technique, did not provide information needed to determine optimal Paw during HFOV in IRDS patients.[9]

We previously showed in a mathematical model that mechanical lung characteristics influence the attenuation of the oscillatory pressure component down the endotracheal tube.[10] To quantify the attenuation, we defined the oscillatory pressure ratio (OPR) as the ratio of pressure swings at the distal and proximal ends of the endotracheal tube. We found an inverse relationship between C_{rs} and OPR: when C_{rs} increased, the attenuation of pressure swings increased and thus OPR decreased. We noted that OPR was high both at low Paw, when the lungs were partly collapsed, and at high Paw, when the lungs were overdistended. OPR was minimal at the maximal C_{rs}, at maximal lung recruitment and minimal overdistention.

The objective of the present study was to test in vivo, firstly whether there is a consistent relationship between OPR and Paw with an identifiable minimum, and secondly whether OPR reaches a minimum at the optimal Paw, i.e., the lowest Paw where $Q'_s/Q'_T < 0.1$. We studied OPR in lavage-induced surfactant deficient pigs, which were subjected to a stepwise increase of Paw to serially impose underinflation, optimal inflation, and overdistention of the lungs, followed by a stepwise decrease of Paw.

ABBREVIATIONS

CMV	Conventional mechanical ventilation
C_{rs}	Respiratory system compliance
F_{IO2}	Inspired oxygen fraction
HFOV	High-frequency oscillatory ventilation
I:E	Inspiratory:expiratory ratio
IRDS	Ideopathic respiratory distress syndrome
OPR	Oscillatory pressure ratio
P_{aCO2}	Arterial carbon dioxide tension
P_{aO2}	Arterial oxygen tension
P_{ao}	Aortic blood pressure
P_{pa}	Pulmonary artery pressure
P_{cv}	Central venous pressure
Paw	Mean airway pressure (on HFOV)
Paw_{minOPR}	Paw at minimal OPR
Paw_{opt}	Optimal Paw
PEEP	Positive end-expiratory pressure
P_{plat}	Airway plateau pressure
Q'_S/Q'_T	Physiological shunt fraction
Q'_T	Cardiac output
T_b	Central blood temperature
V_T	Tidal volume
V_D/V_T	Dead space as a fraction of tidal volume
V_{EE}	End-expiratory lung volume
V_{th}	Thoracic expansion (plethysmograph)
V_A'/Q'	Alveolar ventilation-perfusion ratio
ΔP_{dist}	Pressure swing at distal end of endotracheal tube
ΔP_{prox}	Pressure swing at proximal opening of endotracheal tube

MATERIALS AND METHODS

Surgical Procedures

Eight Yorkshire pigs (12.0 ± 0.5 kg body weight) were anesthetized with sodium pentobarbital ($30\ mg \cdot kg^{-1}$), followed by a continuous infusion ($8.5\ mg \cdot kg^{-1} \cdot hr^{-1}$). The pigs were ventilated through a Y-shaped tracheal cannula in a volume-controlled mode, using a ventilator allowing for lung volume estimation by the open helium wash in method.[11] Ventilatory frequency was 10 breaths per minute, F_{IO2} was 0.21, and PEEP was 5 cm H_2O. Inspiratory:expiratory (I:E) ratio was 1:1, with an inspiratory pause of 0.6 sec. Tidal volume ($V_T = 17.1 \pm 0.9\ ml \cdot kg^{-1}$) was adjusted to achieve normocarbia (P_{aCO2} 38–45 mm Hg). An arterial line was inserted in the aorta for measuring aortic blood pressure (P_{ao}). A pulmonary arterial catheter was used to monitor pulmonary arterial pressure (P_{pa}) and pulmonary blood temperature (T_b) and to sample mixed venous blood. A catheter was inserted into the right atrium for injection of saline during the thermodilution procedures to measure cardiac output (Q'_T). A four-lumen catheter was inserted into the superior vena cava to measure central venous pressure (P_{cv}) and to infuse fluids ($5\ ml \cdot kg^{-1} \cdot hr^{-1}$ saline) and anesthetics. The pressure catheters were flushed ($3\ mL \cdot hr^{-1}$) with normal saline containing 10 IU mL^{-1} heparin. The bladder was catheterized to measure urinary output. A mercury-in-rubber strain gauge was fixed around the thorax 3 cm above the xyphoid. After the initial procedures, pancuronium bromide ($0.3\ mg \cdot kg^{-1} \cdot hr^{-1}$) was given to suppress spontaneous breathing.

Measurements and Lavage Procedure

Baseline measurements were performed during CMV prior to lung lavage (*prelavage measurements*). Arterial and mixed venous blood gas samples were analyzed with ABL3 and OSM2 hemoximeters (Radiometer, Copenhagen, Denmark). Inspiratory and mixed expiratory gases, including helium, were analyzed by a mass spectrometer (MGA 1100, Perkin-Elmer, Pomona, CA). Physiological shunt fraction (Q'_s/Q'_T) and physiological dead space (V_D/V_T) in percentage were calculated according to standard formulas.[12] Respiratory system compliance (C_{rs}) was derived according to $C_{rs} = V_T/(P_{plat} - PEEP)$, with P_{plat} being the plateau pressure. Next, F_{IO2} was set at 1.0 and maintained constant during the experiment. PEEP was set at 2 cm H_2O. Repeated lung lavages were applied with $35\ mL \cdot kg^{-1}$ saline until P_{aO2} was below 80 mm Hg.[13] After lung lavage, baseline measurements were repeated (*postlavage measurements*). In addition, the end-expiratory volume (V_{EE}) of the lungs was estimated using an open helium wash-in method.[11] Thoracic expansion (V_{th}) was determined from changes in electrical

resistance of the mercury-in-rubber strain gauge. V_{th} was calibrated at V_{EE} and by application of three inspiratory pauses at inspiratory volumes of 6, 12, and 18 $mL \cdot kg^{-1}$. Next, a 5.5-mm-diameter endotracheal tube was inserted through the tracheal cannula. Proximal airway pressure (P_{prox}) was measured (Viggo-Spectramed DT-NN, Oxnard, CA) via a side port on a connector inserted between the endotracheal tube and the Y-piece. Distal airway pressure (P_{dist}) was measured via a 3F catheter tip manometer (Dräger, Lübeck, Germany) inserted in the Y-piece and advanced 0.5 cm past the tip of the tube. Both pressure signals were amplified (Hewlett Packard 78834A, modified to pass signals up to 75 Hz) and digitally sampled at 200 Hz for 1 min for off-line analysis. P_{dist} was digitally filtered (8th order Butterworth bandpass filter, 5–30 Hz) prior to peak-to-peak analysis to reduce the influence of noise and baseline variation. HFOV (SensorMedics 3100A, Yorba Linda, CA) was initiated at a frequency of 8 Hz and an inspiratory time of 33%, maintained constant during the experiment. Bias flow was set at 20 L/min. Initial Paw was set 3 cm H_2O higher than the mean airway pressure during CMV. We intended to set proximal pressure amplitude ΔP_{prox} to obtain normocarbia. Since this caused a significant improvement in oxygenation, we chose a smaller amplitude (54 ± 6 cm H_2O), which remained unaltered during the experiment.

After the experiment, the tube was removed and CMV resumed at the initial settings. After a 30-min stabilization period, the postlavage measurements were repeated (*end experiment measurements*) to check stability of the model. The fluid balance was determined. On completion of the experiment, the lungs were examined macroscopically by opening the thorax.

Experimental Procedure

Starting from the baseline value, Paw was increased in 3-cm H_2O steps, until P_{aO2} exceeded 450 mm Hg or the increase in P_{aO2} was $< 50\%$ of the previous increase. Subsequently, one step of Paw was added, and then Paw was set at 40 cm H_2O (*inflation*). Paw was then lowered to the value at which the above P_{aO2} criterion was reached, and was subsequently decreased in 3-cm H_2O steps until P_{aO2} was about 60–80 mm Hg (*deflation*). At each Paw, a 10-min stabilization period was applied before measurements were made.

Data Analysis

Data are presented as means $\pm$ 1 SD. To investigate stability of the model *postlavage* and *end experiment*, measurements were compared using the paired *t*-test. During *inflation* and *deflation*, the optimal Paw (Paw_{opt}) was determined as the lowest Paw where $Q'_s/Q'_T < 0.1$, and the Paw at the minimal OPR (Paw_{minOPR}) was determined. The Paw values were compared using the Wilcoxon signed rank test. $P < 0.05$ was accepted as statistically significant.

RESULTS

Lung lavage produced a large increase of Q'_s/Q'_T, an increase in V_D/V_T, and a decrease of C_{rs}, as is typical for IRDS (Table 1). Comparison of the *postlavage* and *end experiment* measurements (Table 1) showed significant increases in P_{aCO2}, P_{pa}, and heart rate. End-expiratory V_{th} increased by 67%, while V_{EE} decreased slightly. The fluid balance was 54 $mL \cdot kg^{-1}$ positive over a 7.5-hr period. When the thorax was opened after the experiment, a substantial amount of yellow-colored fluid was found in the pleural space in all animals. No animal had a pneumothorax. Visual inspection of the lungs revealed significant hemorrhagic edema in all animals.

During *inflation*, Paw was increased stepwise from 17 ± 2 cm H_2O to 32 ± 3 cm H_2O and subsequently to 40 cm H_2O; *deflation* started at 40 cm H_2O, and then Paw was stepwise decreased from 29 ± 3 cm H_2O to 9 ± 2 cm

TABLE 1— Physiological Variables During Conventional Mechanical Ventilation[1]

	Prelavage	Postlavage	End experiment
P_{aO2}/F_{1O2} (mm Hg)	490 ± 48	60 ± 11	96 ± 61
Q'_s/Q'_T	0.01 ± 0.02	0.46 ± 0.08	0.39 ± 0.14
P_{aCO2} (mm Hg)	42 ± 2	52 ± 5	60 ± 8*
V_D/V_T (%)	40 ± 4	61 ± 4	64 ± 5
V_{EE} ($mL \cdot kg^{-1}$)		8.0 ± 2.5	7.5 ± 2.2
V_{th} ($mL \cdot kg^{-1}$)		8.0 ± 2.5	13.4 ± 5.0*
C_{rs} ($mL \cdot cm\ H_2O^{-1} \cdot kg^{-1}$)	1.20 ± 0.16	0.54 ± 0.06	0.55 ± 0.07
Q'_T ($mL \cdot s^{-1} \cdot kg^{-1}$)	1.9 ± 0.3	1.9 ± 0.4	1.9 ± 0.5
P_{ao} (mm Hg)	90 ± 9	85 ± 12	83 ± 18
P_{pa} (mm Hg)	16 ± 2	26 ± 5	32 ± 6*
P_{cv} (mm Hg)	4.3 ± 0.9	4.2 ± 1.2	3.7 ± 1.1
Heart rate (min^{-1})	171 ± 18	119 ± 16	154 ± 40*

[1]Values at *end experiment* were tested against *postlavage*.
*$P < 0.05$.

H_2O. In the first experiment, oxygenation deteriorated rapidly when Paw was lowered from 10 to 7 cm H_2O; only part of the measurements could be completed before we increased Paw. In the following seven experiments, we performed smaller Paw decrements (1–2 cm H_2O) at the end of the *deflation* phase.

The OPR showed a similar behavior in most of the animals (Fig. 1). During *inflation*, the OPR initially decreased from 0.20 ± 0.05 to a minimum of 0.10 ± 0.01 at $Paw_{minOPR} = 31 \pm 4$ cm H_2O, and subsequently increased to 0.12 ± 0.02 at a Paw of 40 cm H_2O (Table 2). During *deflation*, the OPR decreased to a minimum of 0.04 ± 0.01 at $Paw_{minOPR} = 18 \pm 1$ cm H_2O, and then increased to the final value of 0.31 ± 0.18 at $Paw = 9 \pm 2$ cm H_2O (Table 2). In one animal, a different pattern was found, i.e., an initial rise in OPR during *inflation* (Fig. 1, panel 6).

P_{aCO2} increased substantially during *inflation* from 56 ± 10 to 82 ± 11 mm Hg (Fig. 2). During *deflation*, P_{aCO2} decreased to 43 ± 5 mm Hg at $Paw = 18 \pm 2$ cm H_2O, and remained constant until $Paw = 13 \pm 4$ mm Hg, and in six animals increased again. Final P_{aCO2} was 50 ± 6 mm Hg.

Q'_s/Q'_T decreased from 0.53 ± 0.1 to 0.06 ± 0.03 during *inflation*, and increased to 0.61 ± 0.20 during *deflation* (Fig. 2). The optimal Paw criterion ($Q'_s/Q'_T < 0.1$) was reached in all but one animal (subject 6) during *inflation*. In this case, we decided to take the Paw value at the minimum Q'_s/Q'_T. This led to $Paw_{opt} = 31 \pm 4$ cm H_2O (*inflation*) (Table 2). During *deflation*, the optimal Paw criterion was reached in all animals at $Paw_{opt} = 14 \pm 2$ cm H_2O (Table 2). Comparison of Paw_{minOPR} and Paw_{opt} revealed no significant difference (-0.8 ± 3.9 cm H_2O) during *inflation*, and a significantly higher Paw_{minOPR} (4.1 ± 1.6 cm H_2O) during *deflation*. We compared Q'_s/Q'_T, OPR, and P_{aCO2} to the values at Paw_{minOPR} in the *deflation* phase (Fig. 3). OPR at each Paw step was significantly different from OPR at Paw_{minOPR}. Change in Q'_s/Q'_T reached statistical significance at a Paw 6 cm H_2O below Paw_{minOPR}. Change in P_{aCO2} reached statistical significance at Paw higher than Paw_{minOPR}.

DISCUSSION

The substantial increase in thoracic circumference measured by the strain gauge plethysmograph may have been caused by an increase in thoracic fluid volume, as end-expiratory lung volume did not change significantly. This was confirmed by the finding of accumulated intrapleural fluid postmortem. The hemorrhagic edema was found to a lesser degree in previous experiments, where the same lavage procedure and ventilatory conditions (CMV) were applied.[13] The application of HFOV at high lung volume may have amplified the lung injury by overdistention of the parenchyma. As this did not lead to significant alterations in physiological indices from *postlavage* to *end experiment*, we concluded that the animal model was sufficiently stable.

Both during the inflation and deflation part of the protocol, the predicted OPR pattern[10] was found. The OPR initially decreased and subsequently increased with increasing Paw, and similarly with decreasing Paw. In one animal, a slightly different pattern was found (Fig. 1, panel 6): the OPR initially rose during *inflation*. This OPR increase, however, was accompanied by a progressive decrease in P_{aO2}. We speculate that in this case, a Paw below 28 cm H_2O was insufficient to prevent a slow loss of lung volume, causing a decrease in oxygenation and an increase in OPR secondary to decreasing C_{rs}.

The observed range of OPR values was 0.02–0.53, comparing well with the values found in the literature. Gerstmann et al.[14] measured tracheal pressures in nine normal open-chest adult rabbits during HFOV at varying frequencies, at a Paw of 7 cm H_2O. The OPR was on the order of 0.03. Zobel et al.[15] found an OPR of about 0.11 in lung-lavaged pigs during HFOV at 10 Hz and a mean Paw of 31 cm H_2O. Boynton et al.[16] found OPR values on the order of 0.47 in neonates with IRDS during HFOV at 20 Hz at a Paw of 12 cm H_2O. In the latter study, the imposed Paw may have been insufficient to optimally recruit the lungs and thereby reduce intrapulmonary pressure swings. The study was performed long before the beneficial effect of volume recruitment had been demonstrated.[17] Our observations are also in accordance with those of Sakai et al.[18] who found an OPR of about 0.25 in rabbits of 3.3 kg during HFOV at 15 Hz. They varied Paw and observed a rise in OPR at Paw > 25 cm H_2O in healthy animals, suggesting a relation between overdistention and rise in OPR. In lung-lavaged animals, the OPR rose at Paw < 15 cmH_2O, suggesting an inverse relation between degree of atelectasis and OPR.

During the inflation phase, P_{aCO2} increased substantially with increasing distending pressure. P_{aCO2} subsequently decreased with decreasing Paw in all animals, and increased at low Paw in six animals. A number of mechanisms may be involved in eliciting changes in CO_2 elimination with delivery of distending pressure. Firstly, a rise in Paw will cause an increase in lung volume and in addition an increase in anatomical dead space volume, secondary to longitudinal and radial stretching of the conducting airways. In the presence of the low tidal volumes used during HFOV, this may cause a considerable decrease in CO_2 clearance, as previously demonstrated in dogs by Baum et al.[19] Secondly, when a rise in Paw causes a decrease in respiratory system compliance, tidal volume will decrease at constant pressure swings at the airway opening, leading to a decrease in CO_2 clearance. Thirdly, when an existing ventilation-perfusion (V'/Q') mismatch improves on account of alveolar

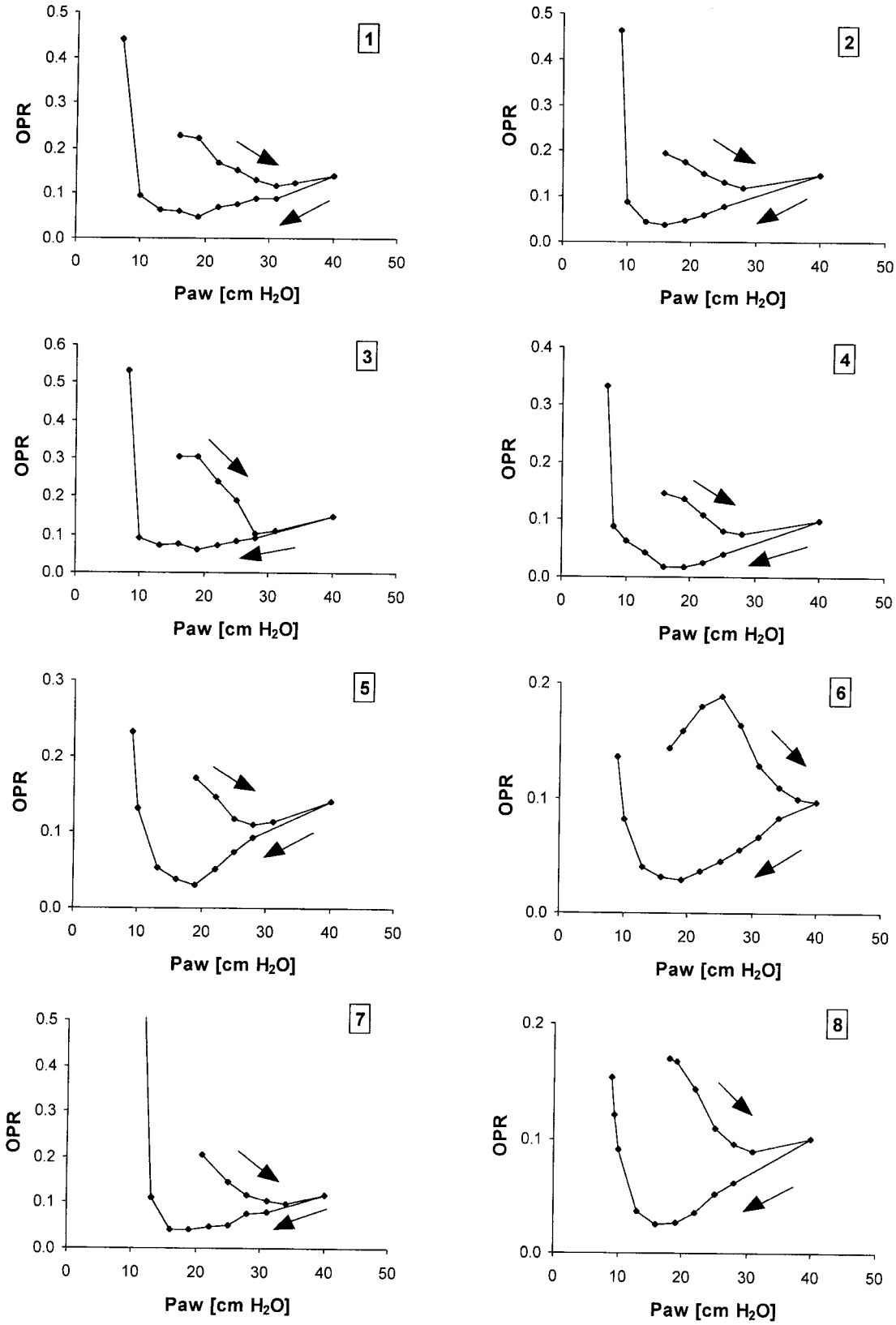


Fig. 1. OPR as a function of Paw during HFOV in all animals (1–8). Arrows indicate order of Paw application.

TABLE 2—Physiologic Variables at Mean Airway Pressure at Which Oscillatory Pressure Ratio Was Minimal and at Optimal Mean Airway Pressure During Inflation and Deflation Phase

	At Paw_{minOPR}	At Paw_{opt}
Inflation		
Paw (cm H_2O)	31 ± 4	32 ± 6
P_{aO2} (mm Hg)	465 ± 54	483 ± 46
OPR	0.10 ± 0.01	0.11 ± 0.02
Q'_s/Q'_T	0.09 ± 0.03	0.08 ± 0.02
Deflation		
Paw (cm H_2O)	18 ± 1	14 ± 2
P_{aO2} (mm Hg)	535 ± 29	509 ± 40
OPR	0.04 ± 0.01	0.04 ± 0.02
Q'_s/Q'_T	0.05 ± 0.02	0.07 ± 0.02

recruitment by increasing Paw, CO_2 elimination improves. However, a good V'/Q' match may deteriorate on account of the decrease of pulmonary perfusion with increasing Paw,[20] leading to decreased CO_2 elimination. The observed P_{aCO2} pattern may be explained by the changes in anatomical dead space volume as well as changes in respiratory system compliance. The rise in P_{aCO2} in most of the animals at low Paw on the *deflation* limb may be explained by the decrease in C_{rs} alone. In our study, we chose not to achieve normocarbia at each Paw step for two reasons. Firstly, OPR is influenced by changes in ΔP_{prox} due to the nonlinearity of endotracheal tube resistance.[10,21] This would have complicated the interpretation of the OPR pattern. Secondly, the length of the experiment would have increased considerably when at each inflation level a second physiologic stabilization period would have been added after adjustment to the desired ΔP_{prox}. This would have endangered the model stability. In clinical practice, however, CO_2 clearance is controlled by regular adjustment of proximal pressure amplitude. The consequence on OPR pattern remains to be investigated.

Significant hysteresis was found in the physiological shunt fraction in all animals. The opening pressure of the lungs appeared to be higher than the closing pressure. In addition to this hysteresis, the OPR was smaller on the *deflation* limb than on the *inflation* limb, in accordance with Sakai et al.[18] We speculate that during *inflation*, lung recruitment took place up to at least 31 cm H_2O, which was on average the lowest Paw where $Q'_s/Q'_T < 0.1$. At this Paw and above, the alveoli in the nondependent part of the lungs may have been overdistended, yielding low C_{rs} and high OPR. The consequence of this assumption is that a recruitment procedure should be as short as possible and should rapidly be followed by reduction of distending pressure to avoid long-term overdistention. On the *deflation* limb, derecruitment starts around 15–20 cm

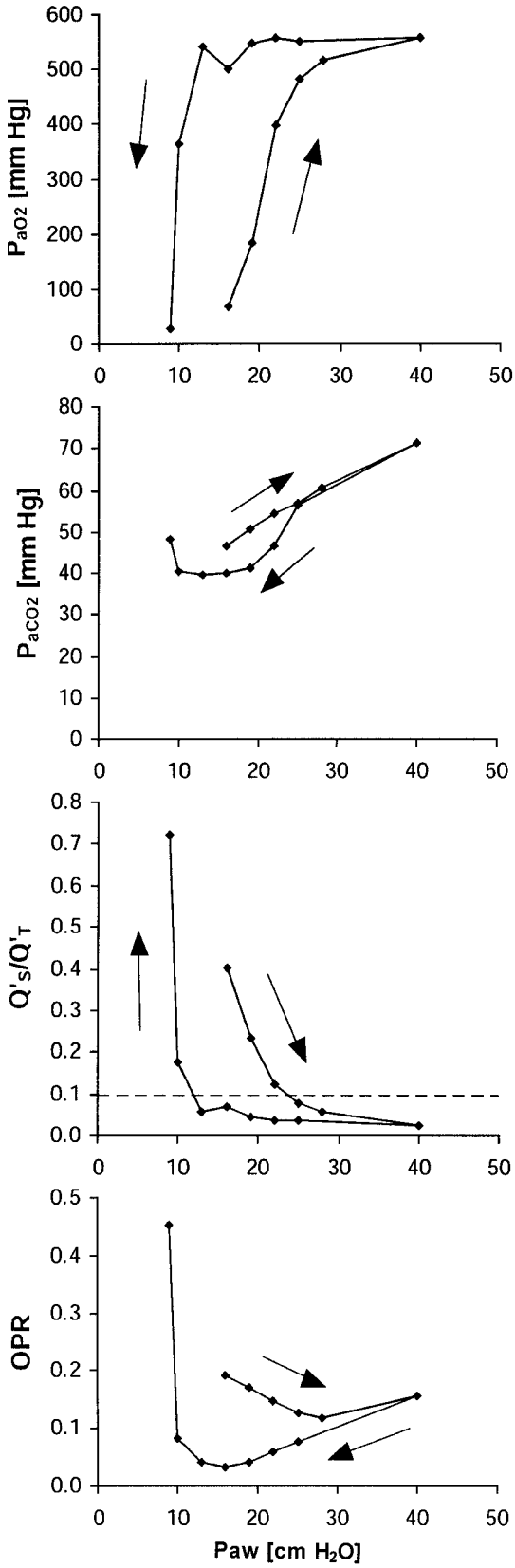


Fig. 2. Relationship between Paw and P_{aO2}, P_{aCO2}, Q'_s/Q'_T, and OPR in one animal. Dashed line is at $Q'_s/Q'_T = 0.1$. Arrows indicate order of Paw application.

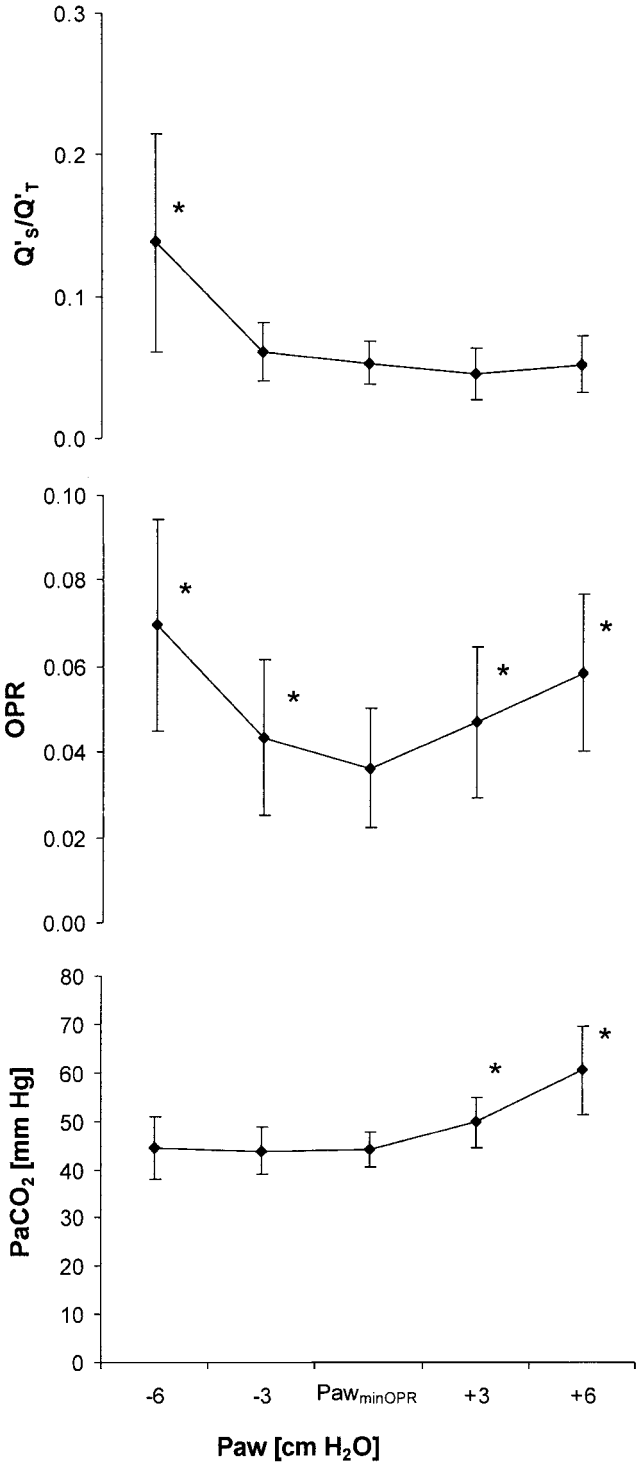


Fig. 3. Q'_s/Q'_T, OPR, and P_{aCO2} as a function of Paw in reference to Paw at minimal OPR (Paw_{minOPR}) in deflation phase of the experiment. Order of Paw application is from right to left. Values were tested against those at Paw_{minOPR}. *$P < 0.05$.

H_2O. At this Paw, alveoli were optimally distended, C_{rs} was larger, and OPR was much smaller than on the *inflation* limb. Thus, the same oxygenation and smaller intrapulmonary pressure swings can be achieved at a lower Paw on the *deflation* limb, following a volume recruitment maneuver than on the *inflation* limb. This may have direct clinical implications, as the lower Paw and smaller pressure swings protect the healthy part of the lungs from excessive distention.

We chose to compare the occurrence of minimal OPR to the optimal Paw to investigate the predictive value of OPR. During *inflation*, Paw_{minOPR} was similar to Paw_{opt}; during the *deflation* phase, Paw_{minOPR} was slightly, but significantly higher than Paw_{opt}. We speculate that, while decreasing Paw, a part of the lungs will become atelectatic, leading to a decrease in C_{rs} and an increase in OPR. This partial lung collapse will be compensated for by regional hypoxic pulmonary vasoconstriction, and thus will hardly cause significant changes in arterial oxygenation. Further decreases in Paw will cause collapse of a larger part of the lungs, a reduction in arterial oxygenation, and an increase in Q'_s/Q'_T. If this speculation is true, the OPR may serve as an indicator of developing atelectasis. Clinical management will preferably take place in the *deflation* phase. A minimum in OPR can only be established after detection of a subsequent OPR increase. The OPR increase occurred well above the Paw when shunt fraction deteriorated dramatically. This may enforce the practical use of the index in clinical management.

In conclusion, in an animal model of experimental lung injury, we found a consistent relationship between OPR and Paw with an optimal minimum for both values. The minimal OPR is reached at a Paw that coincides well with the Paw where oxygenation is optimal. Clinical studies are needed to further clarify the applicability of this index in patients.

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